Recovery-Selection: Learning *Which* Recovery a Robot Policy Should Run

Vignan Kamarthi
RIVeR Lab, Northeastern University

Project Proposal

Thesis. When a learned manipulation policy fails, the best way to recover depends on *what kind* of failure occurred, yet every current system hard-codes a single recovery behavior. I propose to make recovery-strategy *selection* a learnable, cost-aware decision problem, and to show that a small learned selector chooses recoveries that beat any fixed single-mechanism baseline at matched intervention budget.

1. The gap

Recovery from execution failures in robot learning is splintered into siloed lines that each commit to *one* mechanism: retry/rewind (RaC, SPR), local-RL recovery (RecoveryChaining), VLM re-planning (FailSafe, SC-VLA), and human-in-the-loop hand-off (the Cornell HITL framework, HRI 2026). Each works only in the regime it assumes (a rewind needs an in-distribution target; local RL needs a reachable funnel; VLM re-planning fails on perception-OOD states). No published work (verified over four substance-phrased literature probes during a structured search) does all of: (a) formally defines a policy's *competence region* and types failures by their relation to it; (b) learns which of three or more different recovery mechanisms to use, under explicit cost and risk constraints; and (c) evaluates on real-robot rollouts with monolithic end-to-end VLA policies. The nearest neighbor (Cornell HITL) arbitrates only act-versus-ask and only for modular perception/planning/control pipelines, not monolithic VLAs.

2. Contribution

The contribution has three parts:

1. **A failure taxonomy grounded in competence, not error labels.** Each failure is typed by how far the current state lies outside the region where the frozen policy is reliable, rather than by a description of what went wrong. That region is read from two signals: how familiar the state is in the policy's learned representation (latent-state density), and how much a small ensemble of the policy disagrees on the next action, with a control-invariant safe set as a hard safety floor. The state's position then fixes the recovery: in-region → retry; near the boundary → rewind/back-off; outside but plannable → re-plan; outside and risky → ask a human.
2. **A learned selector over the recovery moves.** Given a failure, its features, the competence signals, and the task context, the selector picks one of the five recovery moves, under an explicit budget on how often it may ask a human. This unifies four lines of work that each hard-code a single move.
3. **Counterfactual recovery evaluation.** For each injected failure we reset and replay all five recovery moves and record each one's cost (time, human-intervention effort, safety violations). This yields, per failure, the best possible move (the oracle) and a new metric, *recovery-regret* (the cost paid minus the oracle's cost), which replaces binary recovery-success.

3. Method and plan

A linear, simulation-first workflow:

1. **Make-or-break gate (sim, no robot).** Build the competence model on a frozen VLA (pi0-FAST or OpenVLA, inference/LoRA only). Inject 100–150 scripted failures across 3–4 tabletop tasks; replay all five recovery moves on each failure. **GO** if the per-failure oracle beats the best fixed strategy by $\geq 15\%$ recovery-regret and a linear probe separates failure classes above $\sim$70%. **NO-GO** still yields a publishable benchmark plus metric.
2. **Selector training (sim).** Train the selector on the recorded grid (it learns which move to pick per failure from the measured costs) and calibrate its confidence. Target $\sim$400–600 labeled failure instances.
3. **Real-robot validation (UR3e).** Validate recovery-regret against the four fixed-mechanism incumbents plus random and a hand-coded threshold rule; held-out task generalization; ask-rate/workload curves; ablations on each competence signal.

4. Feasibility and resources

No policy training; only a small selector head and competence calibration, which fit on a single strong GPU such as an H200. The simulation pilot doubles as the training dataset, so the make-or-break gate carries minimal downside before any robot commitment. The one robot-coupled stage needs scripted-failure UR3e time with repeatable resets.

5. Venue

Primary target CoRL 2027 (the robotics home venue), with an ICRA workshop as an on-ramp and arXiv at submission; the work cross-lists naturally to NeurIPS / ICLR. The dataset and the recovery-regret metric stand as defensible contributions even if a learned selector appears elsewhere first.

Key related work (citations to finalize): Back-to-the-Manifold (Reichlin et al., IROS 2022, state-density recovery); control-invariant safe set (Bo et al. 2023); RaC, SPR (retry/rewind); RecoveryChaining (local-RL recovery); FailSafe, SC-VLA (self-correcting VLA, 2024) (VLM re-planning); Banerjee et al., Cornell HITL (HRI 2026); TCoT (AAAI 2026); SO-101 recovery benchmark (recovery-aware evaluation).